

Media Optimization: AWS Cheat Sheet

Baseline 2026-10-07. MediaConvert, S3 and Lambda image processing, CloudFront delivery and a Step Functions VOD pipeline. Quick reference only: verify against the official AWS docs and the full pages.

SERVICES

Need	Service
Transcode files	Elemental MediaConvert
Image variants	S3 + Lambda (Pillow, sharp); on demand: Dynamic Image Transformation for CloudFront
Deliver	S3 + CloudFront (OAC, signed cookies)
Package live / JIT	MediaPackage v2
Live	MediaLive; Amazon IVS (managed, low latency)
Orchestrate	Step Functions + EventBridge

Retired: Elastic Transcoder discontinued 13 Nov 2025 (use MediaConvert). **S3 Object Lambda closed to new customers** (Nov 2025).

MEDIA CONVERT CONCEPTS

- **Job:** inputs + output groups. **Job template:** saved groups, input per job. **Preset:** one output.
- Output groups: File, Apple HLS, DASH ISO, CMAF. HLS group = MPEG-TS segments; CMAF = fmp4 for HLS + DASH.
- **QVBR:** quality level 1-10 (7-9 typical) + MaxBitrate; the managed capped CRF.
- **Automated ABR:** MaxRenditions, Min/MaxAbrBitrate on the group.
- Service role: read uploads, write media; caller needs iam:PassRole.
- Use ClientRequestToken for idempotent creates; UserMetadata carries mediaId.

JOB SKELETON (JSON)

```
{ "Role": "arn:aws:iam::ACCT:role/mc-role",
  "JobTemplate": "bookshelf-hls-ladder",
  "UserMetadata": { "mediaId": "42" },
  "Settings": { "Inputs": [ {
    "FileInput":
"s3://uploads/42/original.mov",
    "AudioSelectors": { "Audio Selector 1":
{ "DefaultSelection": "DEFAULT" } },
    "VideoSelector": { } } ] }
// HLS output: Container M3U8, H_264 {
//   RateControlMode QVBR,
//   QvbrSettings.QvbrQualityLevel 8,
//   MaxBitrate 3000000,
//   GopSize 2, GopSizeUnits SECONDS }
// Covers: FILE_GROUP_SETTINGS, Container
RAW,
// Codec FRAME_CAPTURE,
// FramerateDenominator 10,
// MaxCaptures 6, Quality 80
```

CLI AND EVENTS

```
aws mediaconvert create-job --cli-input-
json file://job.json
aws mediaconvert get-job --id ID
aws mediaconvert create-job-template --cli-
input-json file://t.json
# EventBridge pattern
{ "source": ["aws.mediaconvert"],
  "detail-type":
["MediaConvert Job State Change"],
  "detail": { "status":
["COMPLETE", "ERROR"] } }
```

- Regional endpoint works in current SDKs (no DescribeEndpoints).
- Event detail.userMetadata maps the job back to the media row.
- Step Functions: arn:aws:states:::mediaconvert:createJob.sync waits for the job.

IMAGES: S3 AND LAMBDA

- Separate destination bucket (never write to the trigger bucket: loops). Or prefix filter + other prefix.
- Decode keys with unquote_plus; events are at least once: fixed output keys.
- Pillow in a layer: pip install --platform manylinux2014_x86_64 --only-binary=:all: --target python Pillow.
- sharp: npm install --cpu=x64 --os=linux sharp (arm64 for Graviton).
- 1-2 GB memory, 30-60 s timeout; AVIF in a container image or a worker.
- On demand: restrict sizes or sign URLs; Accept in the cache key.
- BlurHash: pure-Python blurhash in the layer; write a .json next to the cover (tags cannot hold the characters).

CLOUDFRONT PRIVATE VIDEO

- Signed **cookies** for HLS/DASH (prefix policy); signed URLs protect one file only.
- RSA 2048 key: create-public-key, create-key-group, trusted key group on /videos/*.
- Cookies: CloudFront-Policy, CloudFront-Signature, CloudFront-Key-Pair-Id.
- Policy resource https://cdn.example.com/videos/42/*, expiry 2 h.
- CDN on a sibling subdomain; Secure, SameSite; CORS response headers policy for the player; S3 private behind OAC.
- Signing is not DRM.

VOD PIPELINE

1. Client: pre-signed PUT to uploads/.
 2. S3 EventBridge notification → rule → state machine.
 3. Validate Lambda: ffprobe, conditional update AWAITING_UPLOAD → QUEUED; DuplicateEvent → AlreadyHandled.
 4. createJob.sync with timeout and Catch.
 5. BlurHash Lambda (may fail without failing the video).
 6. MarkReady / MarkFailed in DynamoDB.
 7. Serve media/ via CloudFront; alarms on ExecutionsFailed.
- Reference design: Video on Demand on AWS (Step Functions, MediaConvert, DynamoDB, SNS, CloudFront).

OPERATIONS

- IAM least privilege: read uploads/, write media/.
- Lifecycle: expire uploads/, abort incomplete multipart uploads.
- DLQ or on-failure destination for Lambda; SQS between S3 and Lambda for retries.
- Reserved concurrency; MediaConvert queue limits.
- Placeholders: region us-east-1, account 111122223333.